

## INDIVIDUAL REFLECTION

Working on the Hotel Accommodation and Reservation Management System using C++ was a valuable learning experience for me. It helped me understand how the concepts taught in class can be applied to a real-world application. During the development of the project, I contributed to different parts of the system and helped connect important modules such as guest registration, room management, reservations, hotel services, billing, and error handling.

While developing the system, we divided the hotel operations into different classes so that the program would be easier to understand and maintain. We used inheritance to represent different types of rooms and services. Abstract classes and pure virtual functions were used to define common features, while runtime polymorphism was implemented using pointers and virtual functions. We also used virtual inheritance where required to avoid duplication and ambiguity when using multiple inheritance.

One of the main challenges I faced was understanding how the different classes should interact with each other without creating conflicts. I also faced difficulties with function overriding, pointers, checking room availability, and calculating the final bill correctly. Debugging these problems helped me improve my problem-solving skills and taught me that even a small error in a condition, function, or pointer can affect the entire program. Testing the system with different inputs helped us identify and fix these issues.

Overall, this project improved my understanding of important C++ concepts such as classes and objects, encapsulation, inheritance, abstraction, constructors, destructors, pointers, virtual functions, pure virtual functions, runtime polymorphism, and virtual base classes. More importantly, I learned how these concepts can work together to build a complete and practical application. This project also improved my confidence in programming, debugging, and working with object-oriented design.